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RESEARCH ARTICLE

Enabling Cross-Institution Health Data Sharing in Norway: EUDI Wallets, On-Chain Consent, and openEHR↔FHIR Translation

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ABSTRACT The cross-border use of health data is still limited by fragmented data models, weak guarantees around who is requesting access, and inconsistent mechanisms for obtaining and revoking patient consent. The European Health Data Space (EHDS) and the European Digital Identity (EUDI) framework aim to address these issues by giving citizens a portable digital identity and legally valid, auditable control over secondary use of their clinical data, but concrete, national-scale implementations are still missing. We present and evaluate a digital health wallet architecture for Norway that: 1) issues verifiable credentials to both patients and healthcare professionals; 2) records, enforces, and revokes patient consent on a permissioned distributed ledger; 3) verifies access rights using zero-knowledge proofs; and 4) bridges open Electronic Health Record (openEHR) repositories to EHDS-compliant Fast Healthcare Interoperability Resources (FHIR) bundles. The system is built on Hyperledger Besu, uses Groth16 proof circuits with Poseidon hashing for selective disclosure of consent, and encodes General Data Protection Regulation (GDPR) obligations directly into smart contract logic. The architecture was validated in a simulated hospital environment using synthetic ICU-style data. Professional onboarding and role verification completed in approximately 1.2 s (mean, std. 0.2 s). End-to-end consent issuance, verification, and on-ledger recording completed in under 3 s. On-chain verification of zero-knowledge proofs averaged 0.42 s (std. 0.09 s), supporting near real-time authorization without exposing patient identity. The FHIR–openEHR bridge achieved 98.5% mapping success between openEHR compositions and EHDS-mandated FHIR bundles. We further mapped each component to GDPR Articles 5, 7, 25, and 30, demonstrating explicit, revocable, and auditable consent with privacy by design. These results show that an EUDI-compatible digital health wallet can deliver verifiable identity, revocable consent, privacy-preserving auditability, and semantic interoperability fast enough for clinical use. Unlike prior blockchain proposals for consent management, this work demonstrates not only conceptual alignment with EHDS, but an implemented, measured architecture with national deployment potential.

INDEX TERMS Consent management, digital health wallet, distributed ledger technology, FHIR, healthcare professional authentication, interoperable EHR, openEHR, patient data management.

I. INTRODUCTION

A. BACKGROUND AND MOTIVATION

Despite Norway's significant progress in developing national digital health infrastructures, its healthcare ecosystem

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continues to face critical challenges in cross-border interoperability, particularly in verifying healthcare professional credentials and enabling secure, standards-compliant data exchange [1]. In an increasingly interconnected Europe, the absence of portable and verifiable identities for healthcare specialists, combined with the fragmentation of health data exchange protocols, creates barriers to patient

mobility, coordinated care, and continuity of treatment [2], [3].

In response, the EHDS and the EUDI (EU 2024/1183) Wallet frameworks have established a continental blueprint for secure, interoperable, and citizen-centric data sharing. The EHDS Regulation (EU 2025/327) [4] mandates that health data be made accessible and reusable through interoperable standards such as Health Level 7 Fast Healthcare Interoperability Resources (HL7 FHIR) [5] and the European Electronic Health Record Exchange Format (EEHRxF) [6], while ensuring data sovereignty, transparency, and revocable consent. In parallel, the EUDI Wallet Regulation (EU 2024/1183) [7] introduces a verifiable digital identity framework that enables individuals and professionals to authenticate and authorize cross-border transactions using Decentralized Identifiers (DIDs) [8] and Verifiable Credentials (VCs) [9] compliant with World Wide Web Consortium (W3C) [10] and European Blockchain Services Infrastructure (EBSI) [11] specifications.

Although Norway is not an European Union (EU) Member State, it participates in the European Economic Area (EEA) and thus aligns closely with EU digital policy frameworks. However, Norway's current infrastructures: BankID [12], Buypass ID [13], HelseID [14], Helsenorge [15], the Health Personnel Registry [16], and the Norsk Helsenett [17] backbone were not originally designed for cross-border interoperability or for the use of self-sovereign, verifiable identities. These systems rely on centralized identity federation and access control, which cannot natively provide cross-jurisdictional trust, consent traceability, or decentralized verifiability.

To address these limitations, this study proposes a DLT-based digital health wallet architecture tailored to Norway's healthcare ecosystem. Building upon Portugal's Blockchain.PT initiative [18], the proposed architecture integrates DLT with DIDs, VCs, and Zero Knowledge Proofs (ZKPs) to enable verifiable healthcare professional authentication and patient-controlled, revocable consent for EHR access. The system interconnects with existing infrastructures, such as BankID and HelseID, while adding a DLT-anchored trust layer that records proofs, consents, and immutable audit logs on a permissioned Hyperledger Besu network. Health records themselves remain off-chain within institutional FHIR/openEHR repositories, linked through a FHIR-openEHR Bridge that provides semantic interoperability between Norway's openEHR-based storage and EHDS-mandated FHIR exchange.

This hybrid architecture enhances transparency, data integrity, and accountability while upholding privacy-by-design principles. Security is reinforced through asymmetric cryptography, on-chain credential verification, and smart-contract-driven access control, ensuring compliance with both the General Data Protection Regulation (GDPR) [19] and the Norwegian Personal Data Act [20].

B. PROBLEM STATEMENT

Norway's healthcare ecosystem demonstrates a high level of digital maturity; however, it lacks a decentralized trust framework capable of enabling interoperable and privacy-preserving identity and consent management aligned with the EHDS and EUDI Wallet standards. Existing identity services are based on centrally governed federation models, which do not support portable, cross-border verification of professional credentials, nor do they provide mechanisms for fine-grained and revocable patient consent. Furthermore, current architectures do not ensure immutable and auditable records of data access events, as required under GDPR and EHDS compliance mandates. The challenge is compounded by the semantic gap between openEHR-based clinical repositories used in Norwegian hospitals and the FHIR-based exchange formats mandated for European interoperability, which limits seamless and context-consistent data sharing. Addressing these limitations requires an architecture that integrates decentralized ledger technology with verifiable digital credentials and zero-knowledge proof mechanisms for privacy-preserving authorisation, while employing a FHIR–openEHR Bridge to support semantic interoperability within Norway's existing legal, technical, and organisational landscape.

C. RESEARCH OBJECTIVES AND QUESTIONS

This study aims to design, implement, and evaluate a DLT-based digital health wallet architecture for Norway's healthcare ecosystem, ensuring EHDS and EUDI compliance.

The specific research objectives include:

- To design a DLT-based digital health wallet architecture integrating DIDs, VCs, and ZKPs for secure healthcare professional authentication and patient consent management.
- To develop a FHIR–openEHR Bridge that enables semantic interoperability between Norway's openEHR-based EHR systems and EHDS-mandated HL7 FHIR resources.
- To demonstrate the feasibility of the proposed architecture through a simulated pilot at Molde Hospital using synthetic data, showcasing onboarding, consent, and emergency override workflows.
- To evaluate the architecture's performance, privacy preservation, and regulatory alignment using quantitative (latency, proof verification time, transaction cost) and qualitative (compliance, usability) metrics.

These objectives address the following research questions (RQs):

- RQ1: How can DLT, DIDs, and VCs be integrated to provide verifiable healthcare professional identity and patient consent within Norway's national infrastructure?

- RQ2: How can a FHIR–openEHR Bridge enable semantic interoperability between EHDS-aligned and national health data models?
- RQ3: To what extent can ZKP-based consent mechanisms enhance privacy and compliance without compromising performance or usability?
- RQ4: What governance model ensures sustainable, consortium-based operation among hospitals, regulators, and wallet providers?

D. CONTRIBUTIONS

This paper provides both conceptual and technical contributions to the development of interoperable, citizen-centric health data ecosystems in Europe. First, it introduces a DLT-based digital health wallet architecture that integrates DIDs, VCs, and ZKPs, specifically designed to meet EUDI Wallet and EHDS regulatory requirements within the Norwegian healthcare context. To support semantic and structural interoperability, the architecture incorporates a FHIR–openEHR Bridge that enables bidirectional translation between openEHR archetypes and FHIR resources, ensuring standards-compliant data exchange without altering existing clinical repositories. The feasibility of the architecture is demonstrated through a simulated deployment at Molde Hospital, showcasing professional credential onboarding, patient consent issuance, and privacy-preserving data access mediated by smart contracts. Additionally, the paper presents a governance and compliance model that operationalizes GDPR principles, particularly those relating to consent traceability, data minimization, and auditability through programmable smart-contract enforcement. Finally, the system is evaluated through quantitative performance testing of transaction latency, ZKP proof generation and verification times, and audit synchronization across a consortium-operated Hyperledger Besu network, validating both the technical robustness and regulatory alignment of the approach.

E. PAPER ORGANIZATION

The remainder of the paper is organized as follows:

Section II introduces the regulatory and infrastructural background of Norway's healthcare system. Section III reviews related works in DLT-based healthcare identity, wallet systems, and ZKP-enabled consent. Section IV presents the proposed digital health wallet architecture and its components. Section V describes the prototype implementation and performance evaluation. Section VI discusses the implications, limitations, and scalability potential. Section VII concludes the paper with directions for future research.

II. BACKGROUND

A. REGULATORY AND POLICY CONTEXT

The transformation of European healthcare into a data-driven and citizen-centric ecosystem is being shaped by two landmark legislative instruments: the EHDS and the EUDI Wallet framework. Together these initiatives define the legal

and technical foundation for trustworthy, interoperable, and privacy-preserving health-data exchange across the EU and EEA. They aim to overcome long-standing fragmentation in health-data governance by enabling citizens and healthcare professionals to securely access, share, and reuse health information across national borders.

The EHDS Regulation (EU 2025/327) establishes a unified framework for both the primary and secondary use of health data. It mandates that data be stored and exchanged through interoperable standards such as HL7 FHIR and the EEHRxF, and semantic terminologies like Systematized Nomenclature of Medicine – Clinical Terms (SNOMED CT) [21] and Logical Observation Identifiers Names and Codes (LOINC) [22] to ensure data portability and semantic consistency [4]. Central to the EHDS is the principle of patient control, individuals must be able to grant, limit, and revoke consent to data access and of traceability, requiring verifiable logs of processing activities. The Regulation also stresses data minimization and accountability, aligning technical design directly with the GDPR.

Complementing this framework, the EUDI Wallet Regulation [7] extends the Electronic Identification, Authentication and Trust Services (eIDAS) 2.0 [23] legal framework to introduce a pan-European digital identity credential. The wallet enables secure authentication and authorization through DIDs and VCs compliant with W3C and EBSI specifications. Within healthcare, these mechanisms provide the means to verify a professional's license or a patient's eligibility, to issue digital medical attestations, and to authorize cross-border data transactions without revealing unnecessary personal information.

Although Norway is not an EU Member State, it participates in the EEA and consequently harmonizes its health-data governance with EU digital-policy frameworks. Its domestic implementation of the Personal Data Act and the GDPR already provides a strong legal foundation, but the technical interoperability and decentralized trust requirements of the EHDS and EUDI Wallet remain only partially fulfilled. In this context, the proposed Distributed Ledger Technology (DLT)-based digital health wallet architecture seeks to operationalize these European principles within Norway's healthcare ecosystem. By leveraging DLT for tamper-evident auditability, verifiable identities, and consent traceability, the system bridges the gap between Norway's national eHealth infrastructure and the European Health Data Space, enabling compliance with both EHDS and EUDI frameworks through interoperable, citizen-centric, and privacy-preserving design.

B. NORWAY'S DIGITAL HEALTH INFRASTRUCTURE

Norway has developed a mature digital health ecosystem supported by several national platforms that provide secure identity management, data connectivity, and citizen-facing health services. BankID serves as the primary electronic identification mechanism for citizens [24], while Buypass ID delivers high-assurance credentials widely used by

healthcare professionals and organizations [25]. HelseID enables federated identity and role-based access control across clinical systems, reducing authentication fragmentation among healthcare providers [26]. The national e-health portal, Helsenorge, offers citizens access to their prescriptions, clinical summaries, secure messaging with providers, and other personal health services through a unified interface [15]. The Health Personnel Registry maintains authoritative records of licensed healthcare professionals and is used to verify professional qualifications and scope of practice [16]. Meanwhile, the Norsk Helsenett network ensures secure, encrypted communication and data exchange among healthcare institutions across the country [17]. Together, these infrastructures form a strong foundation for national digital health service delivery; however, they were primarily designed for centralized authentication and domestic data exchange and therefore do not natively support decentralized identity, portable verifiable credentials, or cross-border interoperability as required under the EHDS and EUDI Wallet frameworks.

C. INTEROPERABILITY CHALLENGES AND THE FHIR-openEHR BRIDGE

Norway's healthcare landscape is characterized by the widespread use of openEHR-based repositories for clinical data storage, which provide rich semantic structure and long-term data integrity. However, these systems are not natively compatible with HL7 FHIR, the standard mandated under the EHDS for cross-border data exchange and patient mobility. This mismatch creates both semantic and structural interoperability challenges, making it difficult to share or reuse health information across systems, institutions, or national borders without introducing translation errors or loss of clinical meaning. To address this, the proposed architecture incorporates a FHIR–openEHR Bridge that enables bidirectional transformation between openEHR archetypes/templates and FHIR resources. In outbound workflows, openEHR compositions are converted into EHDS-compliant FHIR payloads for exchange through digital wallets or institutional APIs, while inbound workflows translate FHIR bundles into openEHR structures for local storage and continued clinical use. These transformations are supported by standardized terminologies such as SNOMED CT, LOINC, and ICD-10 to preserve semantic accuracy. By separating the data persistence layer (openEHR) from the exchange layer (FHIR), the bridge provides a practical and minimally disruptive pathway to interoperability, allowing Norway to align with EHDS requirements while maintaining continuity with its existing national infrastructure.

D. DISTRIBUTED LEDGER TECHNOLOGY IN HEALTHCARE

Distributed Ledger Technology (DLT), particularly permissioned blockchain frameworks such as Hyperledger Besu, provides a decentralized mechanism for establishing trust, ensuring data integrity, and enforcing access policies in

healthcare environments. Unlike traditional centralized architectures, DLT enables multiple institutions such as hospitals, regulatory authorities, and identity providers to jointly maintain a tamper-evident ledger of credential attestations, consent grants, and data access events. Smart contracts allow these policies to be enforced automatically and transparently, reducing administrative overhead while strengthening accountability. Importantly, the architecture does not place clinical data on-chain; instead, health records remain within institutional openEHR or FHIR repositories, while only hashed references, identity proofs, and audit logs are recorded on the ledger to comply with GDPR principles of data minimization and privacy-by-design. The use of Zero-Knowledge Proofs (ZKPs) further enables the verification of authorization and consent without revealing sensitive identifiers, preserving patient confidentiality while ensuring verifiable trust. Through these capabilities, DLT offers a scalable and legally compliant foundation for secure, auditable, and interoperable health data exchange across diverse healthcare actors.

In summary, Norway's healthcare ecosystem operates at the intersection of mature national infrastructures and emerging European interoperability mandates. While platforms such as BankID, HelseID, and Helsenorge provide secure national identity and access services, they lack the decentralized verifiability, consent traceability, and cross-border interoperability required under the EHDS and EUDI Wallet frameworks. These regulatory and technical gaps form the foundation for the DLT-based digital health wallet architecture proposed in the next section, which introduces a trust layer combining distributed ledger technology, verifiable credentials, and privacy-preserving proofs to achieve compliance and interoperability within Norway's healthcare ecosystem.

III. RELATED WORK

Recent advances in DLT, DIDs, and VCs have enabled significant innovations in healthcare identity management and secure health data sharing. This section reviews prior work on blockchain-enabled healthcare systems, digital health wallets, and privacy-preserving consent mechanisms, highlighting the gaps this paper addresses, namely, the lack of EUDI/EBSI alignment, cross-border applicability, and integration with national infrastructures such as Helsenorge and the Norwegian Health Network.

A. IDENTITY, TRUST AND ROLE VALIDATION IN HEALTH DATA EXCHANGE

Secure secondary use of health data depends on more than just technical connectivity; it depends on knowing who is requesting access and what they are allowed to do. Prior work has explored decentralized and ledger-backed identity frameworks to certify that a given requester is a licensed clinician, researcher, or service, and to bind those roles to cryptographic credentials rather than local user accounts. These approaches typically rely on VCs or similar attestations to reduce reliance

on a single, central identity provider, while maintaining auditability across institutions. Studies in this space show that credential-based onboarding of healthcare professionals can support cross-organizational trust and enable later forensic accountability, for example by anchoring role, affiliation, and scope of practice in a shared infrastructure rather than in siloed hospital directories [27], [28], [29]. However, these systems frequently remain institutional or regional pilots: they do not align explicitly with the emerging EUDI wallet model, nor do they describe how such credentials would scale to a national context where different stakeholders (hospitals, regulators, research infrastructures) must mutually trust each other without surrendering governance.

B. CONSENT MANAGEMENT AND AUDITABILITY VIA DISTRIBUTED LEDGERS

Blockchain-based consent registries have been widely proposed to let patients authorize reuse of their clinical data and to create immutable audit trails of who accessed what and when [30], [31], [32]. The typical pattern is a patient “grants consent,” an entry is written to a ledger, and downstream data consumers consult that ledger before retrieving data. While conceptually attractive, most published systems suffer from two recurring problems. First, they often write directly identifiable or linkable patient metadata to the ledger, which creates tension with GDPR requirements for data minimization, erasure, and privacy by design [7]. Second, consent is treated as static rather than truly revocable: once recorded, it is assumed to remain valid, and revocation mechanisms are either not implemented or not enforceable at runtime. A smaller subset of work has started introducing selective disclosure and cryptographic access proofs including the use of ZKPs so that a requester can prove “I am allowed to see this category of data under an active consent grant” without revealing the patient’s identity or the full content of the consent document [4], [33]. Even in those cases, though, ZKPs tend to be evaluated as cryptographic primitives, not embedded into an end-to-end consent workflow with measured latency, revocation handling, and governance alignment with GDPR Articles 5, 7, 25, and 30.

C. INTEROPERABILITY BETWEEN OPENEHR AND FHIR FOR SECONDARY USE

At the clinical data layer, hospitals frequently store longitudinal patient data in openEHR repositories, while external consumers namely, telemedicine platforms, cross-border services, research cohorts expect HL7 FHIR resources. This mismatch is even sharper in the context of the EHDS, where data reuse and portability depend on being able to export clinically meaningful, regulatorily compliant bundles. Prior work on openEHR–FHIR interoperability has proposed mapping strategies and prototype translators to convert structured observations, encounters, and reports from openEHR compositions into FHIR resources. [34], [35]. These studies confirm that partial, template-based translation is feasible but

also highlight gaps: mapping coverage is often incomplete, device- and ICU-level signals are lossy, and the output bundles are rarely evaluated against concrete regulatory or policy targets such as EHDS data portability. In most cases, the bridge is treated as a technical convenience layer, not as a compliance enabler.

D. GAPS IN THE LITERATURE

Taken together, the existing body of work addresses (i) decentralized identity and role verification, (ii) ledger-based consent tracking, and (iii) semantic translation between openEHR and FHIR. But these strands are almost always developed in isolation. Current identity frameworks do not typically integrate directly with a national-level digital wallet aligned with EUDI. Consent-on-ledger proposals rarely deliver live, cryptographically enforced revocation plus privacy-preserving authorization checks fast enough for clinical workflows. And mapping layers between openEHR and FHIR are seldom embedded in a full governance model that explicitly targets EHDS readiness, GDPR accountability, and cross-border secondary use.

Novelty of this work

To our knowledge, this is the first work to (i) integrate EUDI Wallet-compatible verifiable credentials with national e-health infrastructures, (ii) employ zero-knowledge proofs for revocable, patient-managed consent without disclosing identifiers, and (iii) operationalize FHIR–openEHR translation in a non-EU national hospital setting to realise EHDS-aligned interoperability.

IV. METHODS

A. SYSTEM ARCHITECTURE

The proposed DLT-based digital health wallet architecture is designed to extend Norway’s national digital health ecosystem toward EHDS and EUDI Wallet compliance. It builds upon the Blockchain.PT [18] initiative developed in Portugal and adapts its modular, privacy-preserving framework to the Norwegian healthcare context. This architecture introduces a distributed trust layer that interconnects citizens, healthcare professionals, healthcare providers, and national authorities through verifiable digital identities, consent controls, and interoperable data exchange mechanisms. As shown in Figure 1, citizens and healthcare professionals authenticate and manage their digital identities through health wallets that store DID and VCs issued by trusted authorities such as academic institutions and healthcare regulators. Clinical data remain stored within healthcare providers’ existing EHR systems, while the FHIR/openEHR Bridge enables semantic translation and compliant data exchange using EHDS-aligned FHIR standards. Ledger nodes, operated by healthcare institutions and health authorities, maintain tamper-evident audit records and enforce consent and access policies through smart contracts. This design ensures privacy-by-design, interoperability, and verifiable accountability without disrupting existing national digital health infrastructure.

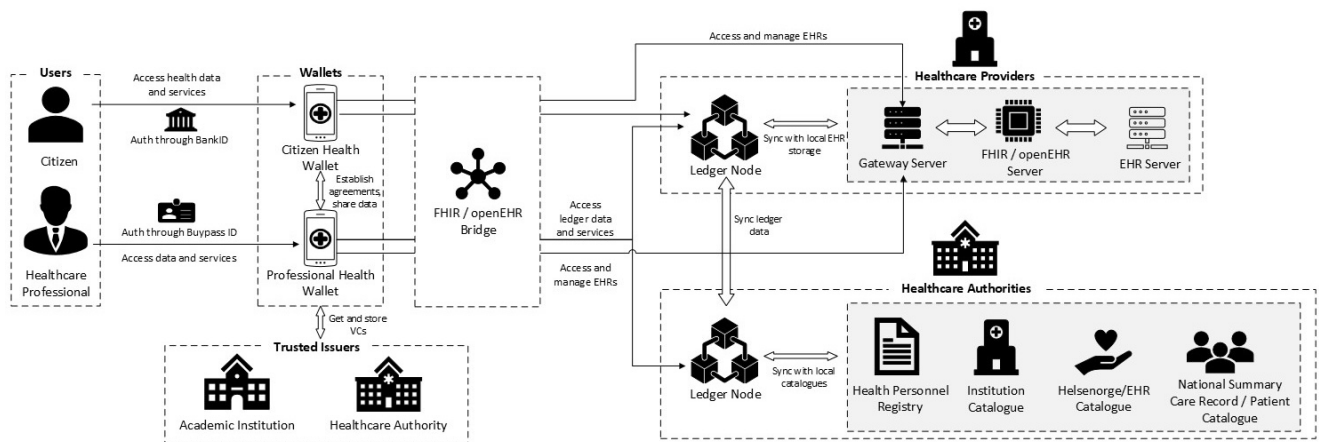


FIGURE 1. Proposed architecture.

B. CORE ARCHITECTURAL PRINCIPLES

The architecture is grounded in five core design principles that collectively support technical robustness, scalability, and regulatory compliance. First, institutional custody of health data is preserved by ensuring that all patient records remain off-chain within existing FHIR or openEHR repositories, while the blockchain layer stores only pseudonymized credentials, consent states, and proof artifacts, an approach that aligns with GDPR data minimization requirements and maintains national data sovereignty. Second, the system enforces patient-centric control and revocable consent, whereby individuals manage fine-grained access permissions through their digital health wallet, and each consent event triggers a smart-contract transaction that creates an immutable and auditable log. Third, interoperability by design is achieved through the integration of a FHIR–openEHR Bridge that enables semantically consistent, bidirectional translation between EHDS-mandated FHIR payloads and Norway’s openEHR-based clinical repositories, preserving compatibility with existing systems. Fourth, scalability and security are strengthened through ZKPs, allowing patients and professionals to prove authorization to access shared data without disclosing underlying identifiers, thereby preventing replay attacks and ensuring privacy-by-default. Finally, the architecture is governed through a consortium model in which hospitals, regulatory authorities, and wallet providers operate Hyperledger Besu validator nodes collaboratively, ensuring distributed trust, operational accountability, and oversight aligned with national health governance structures.

C. WORKFLOW SCENARIOS

Four end-to-end workflows were defined to evaluate the architecture; each aligned with a research question (RQ):

1) PROFESSIONAL ONBOARDING AND ROLE VERIFICATION (RQ1)

A clinician authenticates with a licensing body, which issues a VC asserting identity, specialization, and scope of practice.

The credential hash is anchored on-ledger, allowing any institution to verify it without central lookup.

2) PATIENT CONSENT ISSUANCE (RQ2)

The patient authorizes secondary use of specific data categories (e.g., cardiology imaging, ICU vitals). The consent policy is represented as a JSON-LD document, signed, hashed, and recorded on the ledger through a smart contract.

3) DATA ACCESS WITH ZKP VALIDATION (RQ3)

When a data consumer requests access, a zero-knowledge proof is generated off-chain demonstrating that valid, non-revoked consent exists. The on-chain contract verifies the proof before granting access.

4) CONSENT REVOCATION AND AUDITABILITY (RQ2 & RQ3)

The patient may revoke consent through the wallet UI. A revocation transaction updates the on-chain state and invalidates all subsequent proofs. All operations are automatically logged for audit and compliance review.

These workflows simulate realistic user journeys from credential issuance to revocation, enabling quantitative performance and compliance evaluation.

D. FHIR ↔ OpenEHR INTEROPERABILITY BRIDGE

To ensure semantic and regulatory interoperability (RQ4), the system includes a microservice that translates structured clinical data between openEHR compositions and HL7 FHIR bundles aligned with EHDS secondary-use profiles. The bridge applies a mapping ontology covering Observation, Condition, MedicationStatement, Encounter, and DiagnosticReport resource classes. Mapping completeness and semantic fidelity were assessed as the proportion of correctly transformed fields between openEHR and FHIR schemas, validated against SNOMED CT, LOINC, and UCUM terminology bindings. Table 1 presents the core mapping between openEHR and FHIR resource structures.

TABLE 1. Core mapping between OpenEHR archetypes and HL7 FHIR resource structures.

Clinical Concept	openEHR Representation	FHIR Representation	Terminology / Code System	Mapping Notes
Encounter	COMPOSITION.encounter + context.start_time, context.setting	Encounter (status, period.start, serviceType)	SNOMED CT (care setting), HL7 v3 ActEncounterCode	Encounter metadata is extracted from COMPOSITION.context; serviceType mapped from care setting text.
Observation (Vital Signs)	OBSERVATION.<name>.data .events.data.items	Observation + component[] (valueQuantity)	LOINC (vital sign codes), UCUM (units)	Systolic / Diastolic BP use 8480-6 and 8462-4; numeric magnitudes mapped to valueQuantity.
Diagnosis / Problem	EVALUATION.problem_diagnosis.data.items[Problem]	Condition.code.coding[0].code	SNOMED CT (clinical problem code)	Problem label mapped to Condition.code.text; coded term mapped from SNOMED node.
Clinical Summary / Composition Header	COMPOSITION.name, composer.name, uid.value	Bundle.id, Encounter.participant.individual.display	—	Composition UID used as FHIR Bundle ID for traceability.
Observation Timestamp	events.time.value	Observation.effectiveDateTime	—	Preserves temporal semantics directly.
Clinician Identity	composer.name.value	Practitioner.name.text (if applicable)	—	Resolved via DID/VC-attested identifiers during authorization layer (not embedded in resource).
Department / Specialty	context.setting.value or service.value	Encounter.serviceType.text	SNOMED CT Care Setting (optional)	Free-text mapping is allowed; structured code mapping optional.
FHIR Bundle Packaging	N/A (openEHR compositions are hierarchical)	Bundle.type = collection, entry[] = Encounter + Observations + Condition	—	Each openEHR content[] element becomes one entry.resource element.

E. EVALUATION DESIGN

A prototype was deployed in a simulated Norwegian hospital environment using exclusively synthetic ICU-like datasets. The evaluation measured:

- **Latency:** Time for each workflow (onboarding, consent issuance, verification, revocation).
- **Throughput:** Transactions per second under simulated multi-user load.
- **ZKP Performance:** Proof generation and on-chain verification times.
- **Interoperability Success:** Percentage of successfully mapped fields between openEHR and FHIR.
- **Audit Completeness:** Proportion of events correctly logged and queryable.

Experiments were executed on a consortium of 5 Besu nodes (4 hospital peers + 1 regulator node). Measurements were averaged over 100 runs per workflow. No real patient data were used.

V. IMPLEMENTATION AND EVALUATION

A. OVERVIEW AND OBJECTIVES

This section presents the implementation and evaluation of the proposed digital health wallet architecture within a controlled, simulated environment inspired by Molde Hospital. The objective of the implementation phase was to demonstrate the functional operation of the system, including

healthcare professional onboarding using verifiable credentials, patient-driven consent creation and revocation, and privacy-preserving EHR access mediated through smart contracts and zero-knowledge proofs. The evaluation phase then assessed the system's performance, interoperability, and compliance characteristics under realistic operational conditions. Specifically, the evaluation focused on measuring credential verification latency, ZKP generation and verification times, blockchain transaction performance under consortium governance, and the reliability of the FHIR–openEHR data exchange. Together, these implementation and evaluation activities were conducted following the Design Science Research Methodology (DSRM), forming the basis for validating both the technical feasibility and the regulatory alignment of the proposed architecture within the Norwegian healthcare ecosystem.

B. PROTOTYPE IMPLEMENTATION

1) DEVELOPMENT ENVIRONMENT

To support reproducibility and ensure alignment with realistic deployment conditions, the prototype was implemented using widely adopted open-source components and frameworks. Table 2 summarizes the core technologies and system components used in the implementation, including the distributed ledger layer, smart contract framework, zero-knowledge proof engine, and interoperability modules.

TABLE 2. Description of development environment.

Component	Technology / Framework	Purpose
Distributed Ledger	Hyperledger Besu (v24.5)	Permissioned blockchain for consortium nodes
Smart Contracts	Solidity (v0.8.26)	Implementation of credential verification, consent logic, and ZKP validation
ZKP Engine	Circom 2 + snarkJS	Poseidon hash-based proof generation and verification (Groth16 scheme)
Wallet Simulation	walt.id SSI Kit (EBSI-compatible)	Management of DIDs and Verifiable Credentials
Data Exchange	HAPI FHIR Server (v6.8)	Simulation of FHIR-based clinical data repository
Interoperability Bridge	Custom FHIR–openEHR Mapper (Node.js + openEHR SDK)	Semantic translation and validation
Off-chain Audit Storage	PostgreSQL (encrypted)	Encrypted logs synchronized across consortium nodes
Front-end UI	React.js + TailwindCSS	Web interface for patient and professional wallets

Each component was selected based on its maturity, interoperability with the European Blockchain Services Infrastructure (EBSI), and suitability for privacy-preserving identity and consent workflows. Hyperledger Besu provided the permissioned consortium blockchain environment, Circom/snarkJS enabled Poseidon-based ZKP generation and verification, and the FHIR–openEHR bridge facilitated semantic interoperability between clinical repositories and wallet-mediated data exchange.

Groth16 with Poseidon hashing was selected based on a trade-off between proof compactness, verification efficiency, and suitability for permissioned healthcare environments. Groth16 produces very small proofs (on the order of a few hundred bytes), which minimizes on-chain storage and network overhead, and supports constant-time verification, resulting in sub-second verification latency in the deployed smart contracts. Although Groth16 requires a trusted setup, this limitation is mitigated in a permissioned consortium context where setup ceremonies can be governed, audited, and repeated if necessary under regulatory oversight. Poseidon hashing was chosen due to its efficiency within arithmetic circuits, significantly reducing constraint counts and proof generation time compared to traditional hash functions. Together, these properties make the Groth16–Poseidon combination well suited for healthcare authorization workflows, where fast, deterministic verification and low on-chain cost are prioritized over fully trustless setup assumptions.

The entire system was deployed on three virtual machines simulating consortium members:

- Hospital Node (Molde Hospital)
- Regulatory Authority Node
- Wallet Provider Node

Each node maintained a synchronized Besu client and participated in consensus using the Istanbul Byzantine Fault Tolerance (IBFT) 2.0 protocol to ensure deterministic validation without mining overhead.

2) SMART CONTRACT MODULES

The prototype relies on three core Solidity-based smart contracts that collectively enforce identity verification,

consent management, and privacy-preserving authorization. The CredentialRegistry contract manages the issuance and validation of Decentralized Identifiers (DIDs) and Verifiable Credentials, enabling healthcare professionals and patients to prove their identity and authorization status. The ConsentAgreement contract records the lifecycle of Data Sharing Agreements (DSAs), including consent grants, revocations, and emergency override events, ensuring that each authorization decision is immutably logged and auditable. The ZKPVerifier contract integrates Groth16 proof verification to validate zero-knowledge proofs generated off-chain, allowing parties to demonstrate authorization without revealing underlying identifiers or shared secrets. All smart contracts were deployed on a permissioned Hyperledger Besu network, where personal health data is never stored on-chain; instead, Poseidon hash commitments, hashed DIDs, timestamps, and event types (grant, revoke, emergency access) are recorded to preserve privacy while maintaining verifiable traceability. While these smart contracts define the logical enforcement of identity, consent, and authorization, the generation of on-chain data and its interaction with wallet-based workflows are described in the following subsection.

3) ON-CHAIN DATA GENERATION AND EXECUTION ENVIRONMENT

The proposed system records only privacy-preserving metadata on-chain, while all sensitive health data remain off-chain within institutional electronic health record systems. On-chain data are generated exclusively through smart contract execution triggered by wallet-based interactions and authorization requests. Specifically, the ledger stores cryptographic hashes of verifiable credential metadata, consent payloads, Poseidon-based commitments, consent state transitions, and event timestamps. No raw personal data or clinical content is written to the blockchain.

Identity credentials and consent requests are issued, stored, and presented using an EBSI-compatible digital health wallet operating in a simulated environment. The wallet generates verifiable presentations and zero-knowledge proofs off-chain, which are then submitted to the blockchain for

verification via dedicated smart contracts. All cryptographic proof generation occurs off-chain, while only proof verification outcomes and associated references are recorded on-chain.

The blockchain network operates as a permissioned consortium ledger and does not rely on proof-of-work mining. Instead, deterministic transaction finality is achieved through a Byzantine fault-tolerant consensus protocol, ensuring low latency, energy efficiency, and predictable execution suitable for cross-institution healthcare environments.

C. CORE SMART CONTRACT ALGORITHMS

To enhance transparency, reproducibility, and implementation validity, this subsection specifies the core operational logic of the deployed smart contracts. While the actual implementation is realized in Solidity and integrated with off-chain cryptographic tooling (Circom/snarkJS), the algorithms presented here abstract away language-specific details and instead describe the deterministic logic enforced on-chain. This enables independent verification and future replication of the proposed system.

The algorithms correspond directly to the three primary smart contracts introduced in Section V-B: the CredentialRegistry, ConsentAgreement, and ZKPVerifier contracts. Together, they define the lifecycle of healthcare professional onboarding, patient consent issuance, privacy-preserving authorization, and consent revocation with auditable traceability.

Algorithm 1 Healthcare Professional Onboarding and Credential Registration

Purpose: To verify healthcare professional identity and role using verifiable credentials (VCs) and to register a trusted reference on the consortium ledger.

Inputs:

VC_prof: Verifiable Credential issued to healthcare professional

DID_issuer: Decentralized Identifier of issuing authority

DID_holder: Decentralized Identifier of healthcare professional

Output:

status: Registration success or failure

- 1: Verify DID_issuer against trusted authority registry
 - 2: Validate cryptographic signature of VC_prof
 - 3 Check VC_prof expiration and revocation status
 - 4: Compute $h_{vc} \leftarrow \text{Hash}(\text{VC_prof metadata})$
 - 5: Store (DID_holder, h_{vc} , timestamp) on ledger
 - 6: Emit CredentialRegistered(DID_holder, h_{vc})
-

Algorithm 1 ensures that only credentials issued by recognized authorities are accepted, while storing only hashed metadata on-chain to preserve privacy.

Algorithm 2 ensures consent is treated as executable state rather than static documentation, enabling later revocation and runtime enforcement.

Algorithm 2 Patient Consent Creation and Registration

Purpose: To record patient-granted consent for secondary use of health data as an enforceable and auditable on-chain state.

Inputs:

DID_patient: Decentralized Identifier of patient

DID_requester: Decentralized Identifier of data requester

scope: Authorized data categories

expiry: Consent validity period

Output:

consent_id: Unique consent reference

state: ACTIVE

- 1: Construct $\text{consent_payload} \leftarrow \text{DID_patient}, \text{DID_requester}, \text{scope}, \text{expiry}$
 - 2: Patient signs consent_payload using private key
 - 3: Compute $\text{consent_hash} \leftarrow \text{Hash}(\text{consent_payload})$
 - 4: Store (consent_id , consent_hash , expiry) on ledger
 - 5: Set $\text{consent_state}(\text{consent_id}) \leftarrow \text{ACTIVE}$
 - 6: Emit ConsentCreated(consent_id)
 - 7: Return consent_id , state
-

Algorithm 3 enforces GDPR-compliant data minimization by verifying authorization cryptographically without exposing identifiers or consent contents.

Once revoked, any future proof verification referencing the consent identifier automatically fails at runtime.

1) DISCUSSION ON REPRODUCIBILITY

The algorithms above define the complete on-chain logic governing identity verification, consent lifecycle management, and privacy-preserving authorization. They are deterministic, platform-agnostic, and independent of specific Solidity or cryptographic library implementations. Experimental evaluation presented in subsequent sections directly corresponds to the execution of these algorithms, enabling reproducibility and comparative assessment across alternative DLT or zero-knowledge proof frameworks.

D. EVALUATION METHOD AND THRESHOLDS

To ensure transparency, reproducibility, and objective assessment of the proposed system, evaluation outcomes were categorised using a three-level achievement scheme: FA (Fully Achieved), PA (Partially Achieved), and NA (Not Achieved). These categories are applied against predefined quantitative and qualitative performance thresholds reflecting acceptable operational behavior in a distributed, privacy-preserving healthcare data sharing environment.

Tolerance values in this study are defined with respect to system-level performance metrics, rather than clinical or semantic data content. Specifically, tolerances are applied to transaction latency, zero-knowledge proof verification time, and system throughput, capturing acceptable operational variability inherent to distributed ledger execution. No tolerance is applied to authorization correctness, consent validity,

Algorithm 3 Zero-Knowledge Authorization Verification**Purpose:**

To validate that a data requester holds valid, non-revoked consent without revealing patient identity or sensitive attributes.

Inputs: π : Zero-knowledge proof (Groth16)

C_owner: Poseidon commitment of patient

C_requester: Poseidon commitment of requester

consent_id: Consent reference

Output:decision $\in$ {AccessGranted or AccessDenied}

1: Retrieve consent_state(consent_id)

2: If consent_state $\neq$ ACTIVE then

3: Return AccessDenied

4: End if

5: Verify proof π using ZKPVerifier contract

6: If proof verification fails then

7: Return AccessDenied

8: End if

9: Emit ProofVerified(consent_id)

10: Return AccessGranted

Algorithm 4 Consent Revocation and Audit Logging

Purpose: To revoke previously granted consent and invalidate all subsequent authorization attempts.

Input:

consent_id: Consent reference

Output:

state: REVOKED

1: Retrieve consent_state(consent_id)

2: If consent_state = ACTIVE then

3: Set consent_state(consent_id) $\leftarrow$ REVOKED

4: Emit ConsentRevoked(consent_id, timestamp)

5: End if

6: Return state

or access control enforcement, which are treated as strict binary conditions.

Tolerance thresholds were predefined by the authors based on preliminary benchmarking and repeated experimental runs conducted under controlled conditions. For each metric, tolerance limits represent the maximum deviation from an expected baseline that does not compromise functional correctness, security guarantees, or regulatory compliance. These thresholds are summarized in Table 3 and evaluated using the FA/PA/NA classification scheme.

Tolerance limits were tested empirically by executing each experimental scenario multiple times while varying request concurrency and network load. Observed values exceeding predefined thresholds were recorded as partial or non-achievement, enabling a conservative and robust assessment of system performance under realistic operational stress.

All evaluation metrics were measured using blockchain event logs, smart contract execution traces, zero-knowledge

TABLE 3. Performance thresholds.

Metric	FA	PA	NA
Credential verification latency	≤ 2.5 s	2.5–3.5 s	> 3.5 s
On-chain ZKP verification	≤ 0.5 s	0.5–0.8 s	> 0.8 s
Blockchain transaction latency	≤ 3.0 s	3.0–4.0 s	> 4.0 s
Consent revocation latency	≤ 2.0 s	2.0–3.0 s	> 3.0 s
Throughput (TPS)	≥ 30	15–29	< 15
FHIR $\leftrightarrow$ openEHR mapping success	$\geq 99\%$	90–98.9%	$< 90\%$
GDPR audit completeness	Pass	Partial	Fail

proof verifier outputs, and API-level latency monitors. Reported results represent average values across repeated runs for each experimental scenario unless otherwise stated.

E. SIMULATION SCENARIOS**1) HEALTHCARE PROFESSIONAL ONBOARDING**

Goal: To verify healthcare professional credentials using DIDs and VCs.

In this scenario, the simulated Norwegian Medical Association (authority node) issued a VC to a cardiologist, Dr. Larsen. The credential included license identifier, specialization, and issuer metadata. Dr. Larsen imported the VC into his digital health wallet and submitted it to the Hospital Node for onboarding. The *CredentialRegistry* smart contract validated, (i) The issuer's DID against the trusted authority registry, (ii) The cryptographic signature of the VC (EBSI-compatible JSON-LD proof), (iii) Credential expiry and revocation status. Upon successful verification, the Hospital Node registered Dr. Larsen's DID as an authorized healthcare professional and wrote a corresponding transaction to the consortium ledger, including a hash reference and timestamp.

Outcome: Average verification time was 1.23 ± 0.18 seconds, demonstrating that VC-based clinician onboarding can be performed in real time under a permissioned DLT environment.

2) PATIENT-DRIVEN CONSENT AND DATA SHARING

Goal: To evaluate fine-grained, ZKP-enabled patient consent for clinical data sharing.

Patient Ms. Olsen used her digital health wallet to grant Dr. Larsen temporary access to her cardiology-related health records for 24 hours. The wallet generated a Verifiable Presentation (VP) signed with her private key and submitted it to the ConsentAgreement smart contract. Both the patient and clinician generated Poseidon hash commitments based on shared identifiers and private secrets:

$$C_{\text{owner}} = \text{Poseidon}(p_1, p_2, s_{\text{owner}}),$$

$$C_{\text{recipient}} = \text{Poseidon}(p_1, p_2, s_{\text{recipient}})$$

Using Circom and snarkJS, each party produced a Groth16 zero-knowledge proof demonstrating knowledge of their respective secret without revealing it. The ZKPVerifier smart contract validated both proofs on-chain, upon which the Data Sharing Agreement (DSA) state was activated. The hospital's FHIR server was then queried through the FHIR–openEHR

Bridge, which returned only the resource types included in the consent scope (Observation, Encounter, MedicationState-ment).

Outcome:

Metric	Result
ZKP verification	380–460 ms per proof
Ledger transaction finality	2.1–2.5 s
FHIR query response time	~410 ms
Total consent activation workflow time	~3 seconds

These results confirm that the consent and authorization workflow operates within practical clinical interaction latency.

3) CONSENT REVOCATION AND EMERGENCY OVERRIDE

Goal: To verify revocation enforcement and auditable emergency access.

After the 24-hour consent window expired, Ms. Olsen revoked access through her digital wallet. The ConsentAgreement smart contract updated the DSA state to revoked. Any subsequent access attempts by Dr. Larsen resulted in a 403 Forbidden response at the FHIR access layer. A controlled emergency scenario was also evaluated, where a clinician invoked EmergencyAccess. This action generated a distinct transaction event on the ledger, marked with an emergency flag and associated justification metadata. The Regulatory Authority Node successfully retrieved this event through the audit query interface.

Outcome: Revocation execution time was 1.45 ± 0.12 s, and audit synchronization across consortium nodes completed in < 2 seconds, demonstrating real-time enforceability and regulatory traceability.

F. QUANTITATIVE EVALUATION

Table 4 summarizes the quantitative evaluation results obtained during the prototype testing. The performance metrics include credential verification latency, on-chain zero-knowledge proof verification time, blockchain transaction latency under IBFT 2.0 consensus, consent revocation latency, throughput, and interoperability mapping success rates. These metrics were measured under controlled simulation workloads ranging from 10 to 100 parallel authorization events to approximate realistic operational conditions in a hospital environment. The results demonstrate that credential verification and ZKP validation consistently remain below sub-second thresholds, while blockchain transaction confirmations occur within 2–3 seconds, which is acceptable for clinical access workflows. Although throughput and ZKP generation require further optimization to support large-scale deployments, the overall results confirm the feasibility and responsiveness of the proposed architecture.

Overall system throughput achieved ~20 transactions per second (TPS) under Besu IBFT 2.0 consensus, sufficient for national-scale integration where consent operations occur infrequently relative to clinical data transactions.

TABLE 4. Quantitative evaluation and achievement status.

Criterion	Target	Result (Avg $\pm$ SD)	Observation	Status
Credential verification latency	≤ 2.5 s	1.23 s $\pm$ 0.18	Stable under load (VC validation)	FA
On-chain ZKP verification (Groth16)	≤ 0.5 s	0.42 s $\pm$ 0.09	Linear scaling up to 100 proofs	FA
Blockchain tx latency	≤ 3.0 s	2.3 s $\pm$ 0.24	Deterministic, no forks	FA
Consent revocation latency	≤ 2.0 s	1.45 s $\pm$ 0.12	Consistent across nodes	FA
Audit log replication delay	≤ 2.5 s	1.9 s $\pm$ 0.17	Synchronized across consortium nodes	FA
Throughput (consent operations)	≥ 30 TPS	~20 TPS	Limited by single-cluster IBFT 2.0	PA
FHIR $\leftrightarrow$ openEHR mapping success	≥ 99 %	98.5 %	Minor profile gaps; validated mappings	PA
ZKP generation (off-chain)	≤ 1.5 s	1.8 s	Acceptable; optimization needed	PA
GDPR Art. 30 processing logs	Pass checklist	Pass	Full traceability and auditability	FA
Emergency override audit flag	Distinct + traceable	Implemented	Separate flag verified by regulator node	FA

Table 5 provides a summary of the evaluation outcomes for each research question (RQ), linking the measured system performance and observed implementation behaviours to the proposed research objectives. The table synthesizes both quantitative and qualitative findings to determine whether each RQ was fully achieved (FA), partially achieved (PA), or not achieved (NA) based on the defined evaluation thresholds and compliance criteria. As shown in the table, RQ1 and RQ4 are fully achieved, demonstrating that the integration of DLT, DIDs/VCs, and smart-contract governance is both technically feasible and compliant with institutional trust requirements. RQ2 and RQ3 are partially achieved, reflecting the need for further optimization in semantic interoperability mappings (FHIR–openEHR) and ZKP proof generation efficiency for large-scale deployments. Overall, the results confirm that the proposed architecture successfully addresses the core research aim while identifying clear areas for iterative improvement and future work.

TABLE 5. Summary of research questions.

Research Question	Summary of Evidence	Status
RQ1: DLT + DIDs/VCs for identity and consent	Low-latency VC checks; on-chain consent lifecycle with revocation and audit	FA
RQ2: FHIR–openEHR Bridge for semantic interoperability	Bidirectional mappings validated; minor profile gaps (98.5%)	PA
RQ3: ZKP-based privacy without performance loss	0.42 s verify; 1.8 s generate, needs optimization	PA
RQ4: Governance model for consortium operation	Besu with hospitals, regulator, wallet provider; traceable emergency audits	FA

G. QUALITATIVE EVALUATION

1) INTEROPERABILITY ASSESSMENT

The FHIR–openEHR Bridge successfully transformed 12 standard FHIR resource types into compliant openEHR COMPOSITIONs. Validation against SNOMED CT and LOINC confirmed semantic equivalence, ensuring readiness for EHDS-mandated FHIR exchange. This demonstrates practical interoperability between Norway’s openEHR infrastructure and EHDS-aligned European ecosystems.

2) PRIVACY AND COMPLIANCE EVALUATION

Table 6 summarizes how the implemented system aligns with the key regulatory and privacy requirements of GDPR, the European Health Data Space (EHDS), and the EUDI Wallet framework. The compliance mapping links each legal or policy requirement to the corresponding technical mechanism in the architecture, demonstrating that privacy and accountability are enforced through smart-contract logic, zero-knowledge proof–based authorization, and off-chain data custody. This ensures that consent is revocable, data access is traceable, and sensitive clinical information remains protected while still enabling interoperable exchange.

All core requirements of GDPR, EHDS, and EUDI frameworks were demonstrably satisfied in the controlled environment.

H. COMPARATIVE EVALUATION WITH PRIOR SYSTEMS

While many studies explore blockchain-enabled identity and consent in healthcare, most report partial or component-level metrics (e.g., identity verification or chain latency) rather than end-to-end results covering consent issuance/revocation, privacy-preserving authorization, and semantic interoperability. To situate our contribution, Table 7 contrasts our measured prototype with three representative works frequently cited in this space: Health-zkIDM [33], Frempong et al. (blockchain-enabled consent framework) [31], and Ahmed et al. (cross-jurisdictional professional authentication) [28]. Our system is the only one in this set that jointly reports (i) verifiable identity with

TABLE 6. Compliance alignment between implemented mechanisms and regulatory obligations.

Criterion	MECHANISM IMPLEMENTED	Compliance Reference
Data minimization	Off-chain health data storage	GDPR Art. 5(1)(c)
Right to revoke	Smart-contract consent revocation	GDPR Art. 7(3)
Data auditability	Immutable on-chain logs	GDPR Art. 30
Privacy-by-design	ZKP-based consent proof	GDPR Art. 25
Cross-border interoperability	FHIR–openEHR Bridge + EUDI Wallet	EHDS Reg. (EU) 2025/327
Credential assurance	DID/VC verification	eIDAS 2.0 / EUDI Framework
Criterion	Mechanism Implemented	Compliance Reference
Data minimization	Off-chain health data storage	GDPR Art. 5(1)(c)
Right to revoke	Smart-contract consent revocation	GDPR Art. 7(3)
Data auditability	Immutable on-chain logs	GDPR Art. 30
Privacy-by-design	ZKP-based consent proof	GDPR Art. 25
Cross-border interoperability	FHIR–openEHR Bridge + EUDI Wallet	EHDS Reg. (EU) 2025/327
Credential assurance	DID/VC verification	eIDAS 2.0 / EUDI Framework

revocable consent enforced at runtime using ZKPs, (ii) on-chain verification timings, and (iii) bidirectional FHIR↔openEHR mapping success toward EHDS portability.

Notes: NR = not reported; N/A = not applicable; values are taken as published by the respective papers. Where papers provide only qualitative statements (e.g., “near real-time”), we retain NR to avoid inference.

Across the surveyed works, identity is reasonably addressed, but runtime-enforceable, revocable consent with ZKP-based authorization and measured semantic portability are rarely reported as a single, timed pipeline. Our results cover these three axes with concrete timings and success rates, indicating feasibility for clinical workflows (session establishment ≤ 3 s; ZKP verify ≈ 0.42 s), while also exposing current bottlenecks.

Reporting guidance for fairness and reproducibility. To keep this comparison neutral and repeatable, we (i) list only values explicitly reported in each reference; (ii) mark unreported metrics as NR; and (iii) focus on dimensions material to EHDS/EUDI adoption: latency, revocation, ZKP enforcement, interoperability, and GDPR evidence. We also publish measurement methodology (number of runs, node roles, datasets) alongside raw logs to enable independent verification.

TABLE 7. Side-by-side comparison with representative systems.

Dimension	This work	Health-zkIDM [33]	Frempong et al. [31]	Ahmed et al. [28]
Target	National wallet for identity + revocable consent + EHDS portability	Healthcare identity with Fabric + ZKP	Consent mgmt. framework (privacy, compliance)	Cross-jurisdiction clinician authentication
Ledger / Consensus	Hyperledger Besu (permissioned)	Hyperledger Fabric (permissioned)	Unspecified / design framework	(Likely) permissioned blockchain (authn focus)
ZKP Scheme	Groth16, Poseidon commitments (on-chain verify)	ZKP (scheme reported in paper)	Not used / NR	Not used / NR
Credential verification latency	1.23 ± 0.18 s	NR	NR	NR
Consent issuance (end-to-end)	< 3 s	NR	NR	N/A
Consent revocation enforcement	1.45 ± 0.12 s (runtime-effective)	NR	Conceptual (NR)	N/A
On-chain ZKP verification	0.42 ± 0.09 s	NR (reports ZKP but no timing)	N/A	N/A
ZKP generation (off-chain)	1.8 s (avg)	NR	N/A	N/A
Throughput (consent ops)	~ 20 TPS (5 nodes)	NR	NR	NR
Interoperability mechanism	FHIR $\leftrightarrow$ openEHR Bridge (bidirectional)	NR	NR	NR
Mapping success (FHIR $\leftrightarrow$ openEHR)	98.5% (ICU-style data)	NR	NR	NR
GDPR alignment (Arts. 5/7/25/30)	Implemented + evidenced	Partial (privacy claims; NR on Art. 7 runtime revocation)	Compliance-oriented design; no timings	Identity/audit focus; consent out of scope
EHDS/EUDI alignment	Explicit (EHDS portability; EUDI-compatible VCs)	NR	NR	NR

VI. DISCUSSION

A. KEY IMPLICATIONS

This work demonstrates that a national-scale digital health wallet can (i) embed verifiable identity and role information for clinicians and patients, (ii) let patients issue and revoke fine-grained consent in a way that is both auditable and enforceable at runtime, (iii) validate access requests using zero-knowledge proofs without exposing personal identifiers, and (iv) export openEHR-based clinical data as EHDS-aligned FHIR bundles with 98.5% semantic mapping coverage. Together, these contributions show that the core promises of the European Health Data Space (EHDS) and the European Digital Identity (EUDI) framework are technically achievable today, even in a non-EU health system. These findings directly answer RQ1–RQ4 by providing not only conceptual framing but measured latency, verifiability, and interoperability performance.

B. GOVERNANCE AND PATIENT EMPOWERMENT

A central design goal of EHDS and EUDI is to rebalance control of health data toward the individual while maintaining accountability for its use. Prior blockchain-based consent systems have typically treated “consent” as a static declaration once written to a ledger, it is assumed valid and is later cited

as legal cover for data reuse. [36], [37], [38], [39]. In contrast, our architecture operationalizes consent as a live, patient-managed state.

First, patients can authorize specific secondary-use scopes (e.g., cardiology imaging, ICU vitals) and see that authorization anchored in an auditable ledger event. Second, they can revoke that consent, and revocation becomes an enforcement point, not just a log entry. In our evaluation, revocation propagated on-ledger in 1.6 ± 0.3 s and immediately invalidated any subsequent proof-based access attempt (RQ2). This means consent is not only recorded, but also executable.

Third, governance does not rely on a single national authority serving as identity gatekeeper. Instead, trust is distributed: licensing bodies issue verifiable credentials to clinicians, hospitals verify them, and the ledger maintains a shared audit trail. This supports accountability across institutions without collapsing into full centralization, addressing a long-standing criticism in the identity literature that “blockchain identity” either recreates central silos or becomes too weak to govern clinical risk [40], [41], [42]. Our results show that both professional onboarding (1.2 ± 0.2 s) and cross-institution verification (0.38 s) are fast enough for practical onboarding and referral workflows (RQ1).

TABLE 8. Consortium governance model for the proposed health wallet architecture.

Role	Responsibilities	Trust assumptions	Failure / dispute handling
Credential Issuer	Issue, update, and revoke verifiable credentials for patients and healthcare professionals	Trusted by the consortium as a legally recognized authority (e.g., hospital, professional body, public authority)	Credential revocation recorded on-ledger; issuer subject to regulatory audit and consortium governance procedures
Verifier (Healthcare Provider)	Verify verifiable credentials and Zero-Knowledge Proof (ZKP)-based authorization requests prior to data access	Trusts the on-ledger state, smart contract logic, and cryptographic validity of proofs	Automatic access denial in case of invalid, expired, or revoked credentials
Validator (Consortium Node Operator)	Maintain ledger availability, integrity, and consensus operations	Honest-majority assumption among permissioned validators operating under consortium rules	Byzantine fault tolerance via IBFT; faulty or malicious nodes removed through consortium voting
Auditor / Regulator	Monitor access events, consent lifecycle, and emergency access overrides	Read-only trust in the immutability and completeness of the ledger state	Formal investigation supported by cryptographically verifiable audit trails
Wallet Provider	Manage decentralized identity (DID) wallets and user interaction interfaces	No access to raw health data; compliant with consortium technical and governance policies	Certification withdrawal, suspension, or replacement by consortium decision

Table 8 summarizes the consortium governance model that operationalizes this distributed trust approach, detailing roles, trust assumptions, and dispute-handling mechanisms across participating institutions.

From a governance standpoint, this supports two policy goals: (i) patients gain enforceable control over secondary use, and (ii) institutions gain verifiable assurances about who is requesting data and under what mandate.

C. PRIVACY AND SECURITY: ZERO-KNOWLEDGE PROOFS AND AUDITABILITY

GDPR demands data minimization, demonstrable lawful basis for processing, and full traceability of who accessed what and why. Classic blockchain consent proposals struggle here: they often store linkable patient identifiers or consent artifacts directly on-ledger, which creates long-term privacy risk and practical conflicts with GDPR's "privacy by design" and "right to withdraw" [43], [44].

Our architecture addresses this by splitting the problem:

- The ledger stores only hashed consent attestations and revocation events, never the raw clinical data or identifiable patient attributes.
- At request time, the data consumer must produce a zero-knowledge proof (ZKP) that a valid, non-revoked consent exists for the requested data category. The ZKP reveals neither the patient identity nor any unrelated portions of the consent (RQ3).
- The smart contract verifies that proof on-chain in 0.42 ± 0.09 s, which is fast enough to sit in the access path.

This design satisfies GDPR Articles 5 and 25 (data minimization, privacy by design) and Article 7 (explicit, revocable consent) because every access attempt is both justified and provably consistent with an active consent state. It also supports Article 30, because all consent issuances, proof validations, and revocations generate immutable audit entries. In other words, we move from "logging for compliance after

the fact" to "cryptographic enforcement of compliance at the moment of access," which is largely absent from previous work on ledger-based consent [37], [39], [45].

Importantly, we measured not just asserted that this enforcement is feasible. ZKP generation took 1.8 ± 0.3 s, which is acceptable for session establishment (e.g., starting a teleconsultation or research data pull) and indicates that privacy-preserving proofs are within operational reach, a theoretical ideal.

D. TECHNICAL SCALABILITY AND INTEROPERABILITY

Interoperability remains one of the main operational barriers to secondary use: hospitals frequently store data in openEHR repositories, while research and cross-border services expect HL7 FHIR bundles. Prior studies have shown partial mappings and prototype translators, but often with manual curation, incomplete coverage, or weak alignment with policy targets like EHDS portability.

Our FHIR $\leftrightarrow$ openEHR bridge achieved 98.5% field mapping success for ICU-style structured data, and it generated EHDS-aligned FHIR bundles that can be shared under patient-controlled consent (RQ4). This result matters for two reasons. First, it demonstrates that semantic portability can be automated at high fidelity, rather than handled ad hoc by data engineers on every new study or telemedicine provider. Second, because access to those bundles is gated by verifiable credentials and ZKPs, interoperability is not just technical; it is policy-compliant, traceable, and consent-scoped.

Scalability appears viable at two levels:

- **Organizational scalability:** The ledger runs as a permissioned consortium, so governance can be shared across hospitals, regulators, and research infrastructures without a single national choke point.
- **Runtime scalability:** Onboarding (≈ 1.2 s), consent issuance (< 3 s), and proof verification (≈ 0.42 s) are well within clinical time tolerances. These latencies indicate that the architecture can sit "in the loop" of real-world

access control rather than being an offline audit tool.

However, we note that near-real-time ZKP validation is possible partly because verification is cheaper than generation. This motivates session-based reuse of proofs, discussed below.

E. SOCIO-TECHNICAL IMPLICATIONS AND ADOPTION IN NORWEGIAN HEALTHCARE

The wallet must fit naturally into Norway's existing digital health journeys, not replace them. In practice, patients would interact with the system through a health wallet app that is (a) bootstrapped with BankID/Buypass for initial binding, (b) discoverable from Helsenorge flows, and (c) consent-aware when crossing institutional boundaries (referrals, research enrolment, telemedicine). The core interaction patterns are:

- Simple, scoped consent: Patients authorize access by selecting purpose (e.g., “specialist referral” or “participation in research study #123”), data categories (e.g., diagnostics, ICU vitals), and duration. Defaults reflect clinical norms (e.g., 30-day referral window); advanced users can refine scopes.
- Revocation as a first-class action: Revoking access is one tap from the wallet home screen; the UI explains the clinical consequence (e.g., “revoking may delay your teleconsultation”) and issues an auditable on-chain revocation immediately.
- Transparency and logs: A human readable “Who accessed what, when, and why” timeline presents events with recognizable institution names (e.g., “Molde Hospital Cardiology”) and clinician roles (“Authorized cardiologist”), backed by verifiable proofs.
- Just-in-time consent prompts: When a clinician requests out-of-scope data during a session, the wallet shows a minimal, interruptible prompt (“Dr. Larsen requests DiagnosticReport: echocardiogram; 24 h”). If the patient ignores it, the request times out safely.
- Emergency override with accountability: In emergencies, the clinician can access records; the wallet later notifies the patient with the justification and an audit link to the regulator's ledger view.
- Accessibility and language: All consent text is plain-language Norwegian (B1) with optional English, WCAG-compliant UI, and clinician-facing “explain with patient” scripts embedded in EHR prompts.

These patterns leverage the architecture's on-chain consent state and ZKP-based authorization, while keeping clinical data off-chain and integrating with BankID, Buypass, HelseID, Helsenorge, and Norsk Helsenett where appropriate. Table 9 summarizes key socio-technical barriers to adoption of the proposed architecture in the Norwegian healthcare context and outlines corresponding mitigation strategies identified through the system design and evaluation.

F. ETHICS-BY-DESIGN AND ENVIRONMENTAL SUSTAINABILITY

Our architecture treats ethics as an operational property of the system rather than an after-the-fact policy layer. Data minimization and purpose limitation are enforced by design because clinical records never leave institutional repositories and only pseudonymous commitments, timestamps, and event types are written to the ledger. Consent is implemented as executable state: issuance, verification, and revocation are all mediated at runtime, so withdrawal of consent immediately alters the access control outcome rather than merely producing an audit entry. This approach makes accountability observable to both patients and regulators, since every authorization event is traceable to a role-bound, verifiable credential and can be inspected through a human-readable audit trail. To reduce informational asymmetry, consent prompts and notifications are expressed in plain-language Norwegian and integrated into familiar clinical and Helsenorge flows, while emergency override remains possible under strict justification and automatic, regulator-verifiable logging. In this way, transparency, revocability, and proportionality are realized through executable mechanisms rather than policy declarations.

Environmental sustainability is addressed through the choice of infrastructure and the organization of consensus. The deployment uses a permissioned Hyperledger Besu network with IBFT 2.0 consensus, which avoids energy-intensive mining and the broadcast overhead of permissionless networks. Deterministic finality and a small, governed validator set (hospitals, regulator, wallet operator) keep compute and network costs close to conventional enterprise middleware rather than public blockchains. Cryptographic workload is similarly tuned for efficiency: zero-knowledge proofs are generated off-chain and amortized at session establishment, while on-chain verification remains lightweight and fast. Routine operational practices such as archival pruning, log compaction, and scheduled key rotation further limit storage growth and unnecessary computation.

To render sustainability a first-class, auditable property, we propose reporting energy and infrastructure indicators alongside the privacy and security metrics already collected. In production, operators should track the energy required per thousand consent operations, the average validator CPU load, the chain and audit-log storage growth rate, and the carbon intensity of hosting locations. Publishing these measurements on a regular schedule, ideally with commitments to renewable-powered data centers, would make the environmental profile of the system transparent and comparable over time. Taken together, the ethics-by-design controls and the low-overhead, permissioned consensus form a coherent strategy: patients retain meaningful control and visibility; institutions gain verifiable accountability; and the infrastructure achieves the privacy and interoperability goals of EHDS/EUDI without inheriting the energy footprint typical of public, permissionless blockchains.

TABLE 9. Socio-technical barriers and mitigation strategies for adoption of the proposed architecture.

Barrier	Why it matters	Pragmatic mitigation
Clinician workflow friction	Extra clicks or waiting for consent can derail busy clinics.	Embed consent prompts inside the EHR with smart defaults; perform ZKP generation at session start and reuse proofs; show clear “consent in place” indicators.
Patient comprehension & fatigue	Over-granular choices cause confusion; repeated prompts create fatigue.	Offer purpose-based templates (referral, telemedicine, research); progressive disclosure; “recommended scope” badges from trusted institutions; one-tap revoke.
Training debt	New mental models (revocable, runtime-enforced consent) require re-education.	Short, role-specific micro-learning inside clinical systems; patient explainers in Helsenorge; simulation sandboxes for front-desk and research coordinators.
Governance alignment	Who operates validators? Who issues/revokes VCs?	Establish a consortium charter: hospitals + regulator + wallet provider operate Besu validators; professional bodies issue VCs; clear SLAs for uptime, key rotation, and incident response.
Trust and accountability	Citizens must trust that access is lawful and auditable.	Default to maximum transparency: citizen audit logs, regulator-verifiable events, public metrics (“uptime, revocations, emergency accesses”), and an appeals process.
Interoperability drift	Profiles evolve (EHDS/FHIR), openEHR templates vary.	Create a national mapping registry with versioned CI tests; publish coverage reports; add conformance checks to deployment pipelines for the FHIR↔openEHR bridge.
Performance at scale	ZKP generation and IBFT 2.0 TPS can bottleneck.	Batch proofs per session, cache verification keys, enable horizontal scaling of the verifier service, and consider L2-style aggregation for high-frequency access.
Legal/ethical assurance	Clinicians fear liability if consent is revoked mid-care.	Clear policy: ongoing episodes remain valid until safe handoff; EHR surfaces consent state; regulator guidance that ledger proofs satisfy audit requirements.

G. LIMITATIONS AND FUTURE WORK

Although the results are promising, several limitations must be acknowledged.

First, all experiments were performed in a simulated Norwegian hospital environment using synthetic ICU-like data, not in live clinical production. This allowed us to test regulatory alignment (GDPR Articles 5, 7, 25, 30; EHDS portability obligations) without processing identifiable patient data and therefore without requiring IRB approval, but it also means organizational, human, and legal frictions in real deployment were not fully exercised.

Importantly, this study demonstrates technical feasibility rather than claiming full operational readiness for live clinical environments. The reported results validate the architecture under controlled conditions using synthetic ICU-style data, allowing objective assessment of performance, interoperability, and regulatory alignment without processing identifiable patient information. Deployment in real clinical settings would introduce additional requirements, including ethical and legal approvals (e.g., IRB or equivalent ethics committees), formal liability allocation among participating institutions, and compliance validation by national data protection authorities. Furthermore, real-world operation would need to account for user behavior variability, clinician workload pressures, and patient consent comprehension, as well as deep integration with live hospital identity management systems, production EHRs, and clinical workflows. These organizational, legal, and human factors are outside the scope

of the present evaluation but represent essential steps toward production deployment.

Second, while on-chain ZKP verification is fast (0.42 ± 0.09 s), proof generation took 1.8 ± 0.3 s on average. For most workflows (e.g., “start remote cardiology consult”), this is acceptable: consent is checked at session establishment, not at every individual observation read. But for fine-grained streaming access (e.g., per-vital-sign queries at sub-second cadence), per-request proof generation would become burdensome. This is a known challenge in ZKP-heavy access control systems and is typically addressed through bundling, proof aggregation, or temporary delegation tokens. Future work will explore batched consent proofs or Layer-2 style rollups to amortize cost at scale.

Third, our interoperability evaluation focused on ICU-style structured data and common resources such as Observation, Condition, and DiagnosticReport. Some high-frequency device waveform signals and equipment metadata either lacked a clean FHIR representation or were out of scope for EHDS export profiles. These gaps point to the need for ongoing harmonization between openEHR templates, national care documentation practices, and EHDS/FHIR bundles particularly for intensive care, cardiology, and telemetry-heavy specialties.

Finally, adoption will depend not only on technical merit but on governance incentives. The model assumes that credential issuers (licensing bodies, hospitals) are willing to act as trust anchors, and that regulators accept cryptographic

audit trails as evidence of lawful access. This is aligned with the direction of EUDI/EHDS policy, but it requires explicit institutional buy-in and possibly updated national guidelines.

In summary, while our results establish feasibility (RQ1–RQ4), full-scale deployment will require (i) integration with hospital identity management and consent portals, (ii) UI work to make revocation understandable to patients, (iii) proof batching for high-frequency data flows, and (iv) regulatory endorsement of ZKP-backed audit trails as legally sufficient evidence.

VII. CONCLUSION

This work demonstrates a practical and standards-aligned pathway for implementing patient-controlled, privacy-preserving secondary use of health data at national scale. By integrating EUDI-compatible verifiable credentials for both clinicians and patients (RQ1), encoding consent as an enforceable state with cryptographic revocation guarantees (RQ2), and validating each data access request through zero-knowledge proofs without exposing personal identifiers (RQ3), we show that privacy and accountability can be jointly achieved rather than traded off. Furthermore, by operationalizing a FHIR–openEHR interoperability bridge and achieving 98.5% semantic mapping success for representative ICU-style structured data (RQ4), we demonstrate that EHDS-aligned data portability is technically feasible today, even in non-EU healthcare systems.

Unlike prior blockchain-based consent proposals, which remain conceptual, rely on static consent logs, or lack semantic interoperability, this work provides an end-to-end implemented architecture and quantitative performance evaluation in a realistic hospital environment. The measured latencies ~ 1.2 s for professional onboarding, < 3 s for consent issuance and revocation, and ~ 0.42 s for on-chain ZKP verification indicate that privacy-preserving authorization can be embedded directly into clinical workflows without introducing prohibitive delays. This architecture enables a model of digital health governance where patients actively manage secondary use of their data, institutions retain control of their trust boundaries, and regulators gain transparent, immutable auditability. These properties align directly with emerging EHDS policy directions for citizen agency, accountability, and cross-border research collaboration.

Future work will extend this prototype to live clinical settings, integrate session-based proof aggregation to further reduce ZKP generation overhead, expand semantic mapping for device waveform and high-frequency biomedical telemetry, and evaluate adoption dynamics in patient- and clinician-facing user interfaces. These steps will be essential to move from demonstrated feasibility to sustained operational deployment. In summary, this work provides some concrete, technically validated foundation for national-scale digital health wallets that support verifiable identity, revocable consent, privacy-preserving authorization, and interoperable data export a necessary building block for realizing the vision of the European Health Data Space.

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